

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1521

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Code of Conduct

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

I D.G.RAGHAVI (192521173)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

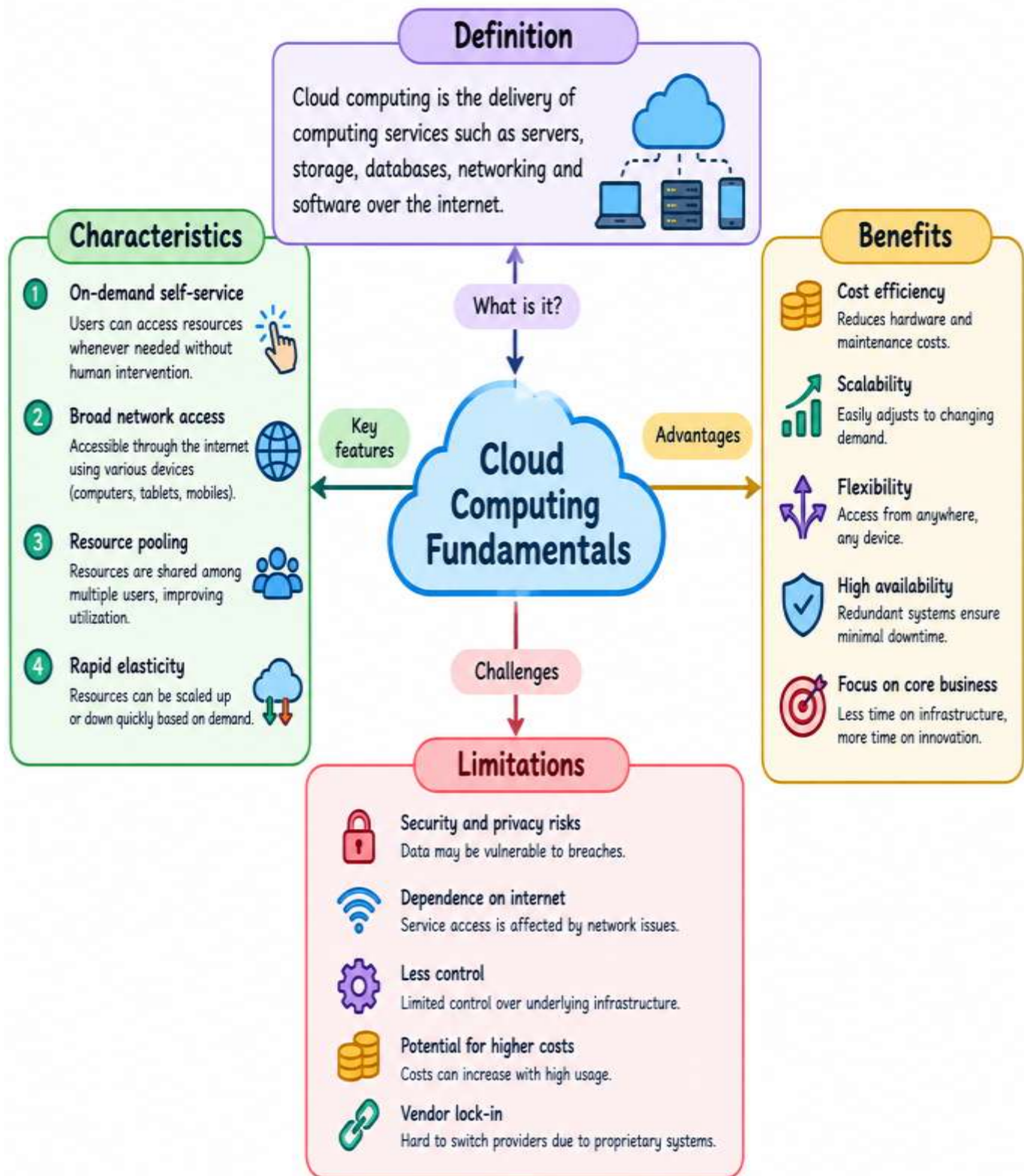
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.G.RAGHAVI

Assessment Tasks Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

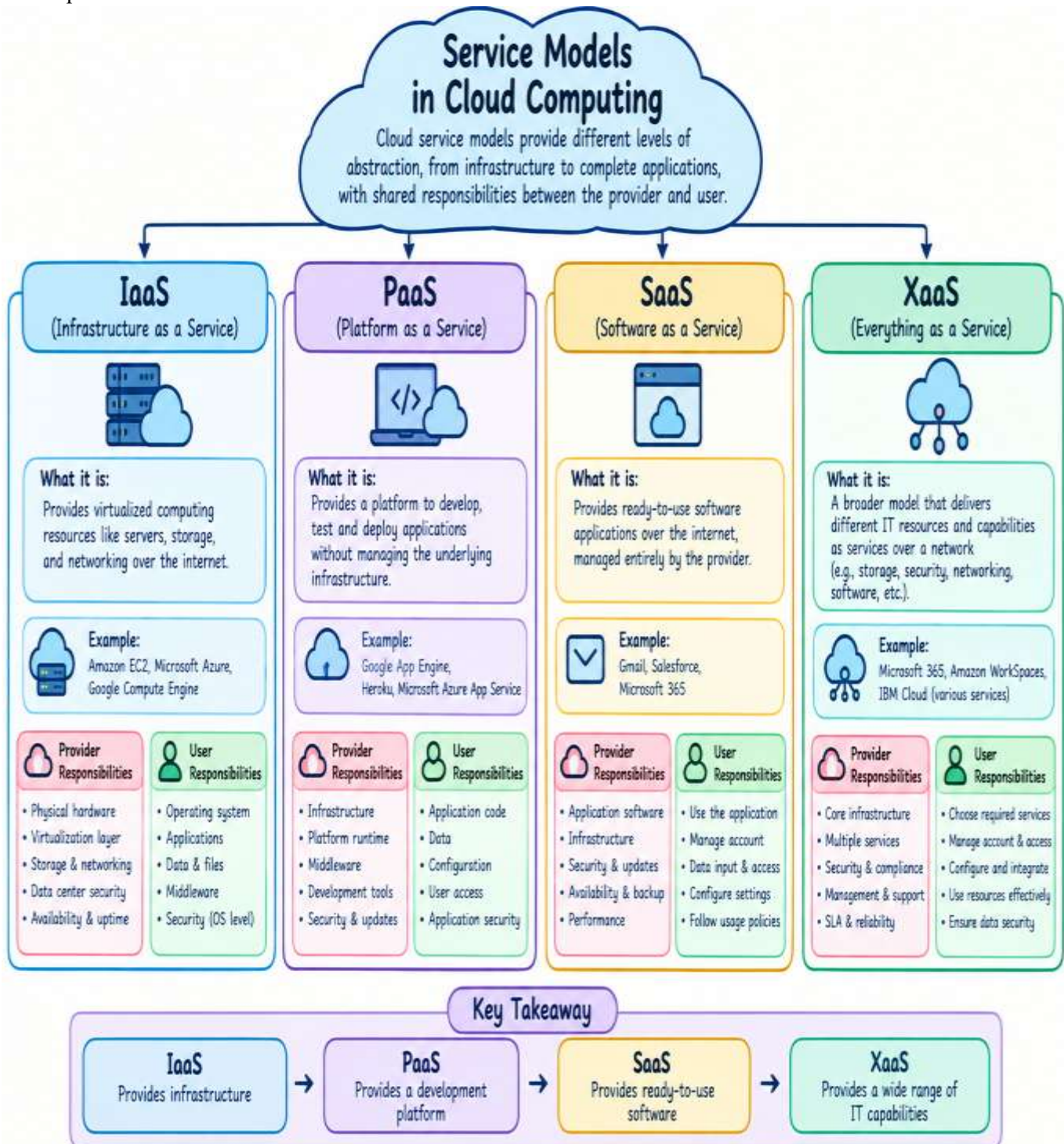
1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.



2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Virtualization and Web Services as Enablers of Cloud

Together, they provide the technology and communication foundation for cloud deployment and scalability.

Virtualization

A technology that creates virtual versions of computing resources (e.g., servers, storage, networks).



Key Features

- Multiple VMs on one physical server
- Better resource utilization
- Isolation and security
- Easy management and migration

How it helps Cloud

- Creates flexible, on-demand resources
- Reduces hardware cost
- Enables multi-tenancy
- Supports rapid provisioning

Web Services

A set of standards-based technologies (e.g., REST, SOAP) that allow applications to communicate over the internet.



Key Features

- Platform independent
- Uses standard protocols (HTTP/HTTPS)
- Interoperability
- Self-describing (APIs)

How it helps Cloud

- Connects distributed services
- Enables communication between applications and data
- Supports hybrid and multi-cloud
- Allows integration with third-party services

Cloud Computing

Provides underlying infrastructure

Enables communication and integration

Supports Cloud Deployment



- Virtualization enables creation of virtual machines and containers.
- Web services provide APIs for managing and accessing resources.
- Together they allow quick and flexible deployment across data centers or regions.

Enables Scalability



- Virtualization allows dynamic allocation of resources (scale up/down).
- Web services enable applications to automatically scale and balance load.
- Combined, they support elastic, on-demand scaling in the cloud.

Result



Virtualization and Web Services work together to make cloud computing flexible, reliable, and easily scalable, supporting efficient deployment and seamless growth of applications and services.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented